

Method 3: HOMOGENEOUS EQUATIONS

(Lesson 21)

Sometimes we can make the inseparable differential equation separable by using substitution eg $y = vx$ or $y = \frac{v}{x}$ or $y = x + v$ etc.

Example 1

Find the solution, of the differential equation

$$\frac{dy}{dx} = \frac{x+3y}{2x}, \quad \text{HINT: let } y = vx$$

not separable

Solution

Step 1: differentiate the given substitution:

$$y = vx \xrightarrow{\frac{d}{dx}} \frac{dy}{dx} = v + x \frac{dv}{dx} \quad (\text{product rule})$$

Step 2: Substitute in the d.e.

$$\frac{dy}{dx} = \frac{x+3y}{2x} \iff v + x \frac{dv}{dx} = \frac{x+3vx}{2x}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1+3v}{2} = \frac{1}{2} + \frac{3v}{2}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1}{2} + \frac{3v}{2} - v = \frac{1}{2} + \frac{1}{2}v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v}{2} \quad (\text{separable})$$

$$\Rightarrow \int \frac{2}{1+v} dv = \int \frac{1}{x} dx$$

$$\Rightarrow 2 \ln(1+v) = \ln x + \ln A = \ln(Ax)$$

$$\Rightarrow (1+v)^2 = Ax, \quad \text{but } \begin{matrix} y = vx \\ v = \frac{y}{x} \end{matrix}$$

$$\Rightarrow \left(1 + \frac{y}{x}\right)^2 = Ax$$

①

Example 2

-2-

Solve $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$

Not Separable

Solution

$$y = Vx \xrightarrow{\frac{d}{dx}} \frac{dy}{dx} = V + x \frac{dV}{dx}$$

Substitute:

$$V + x \frac{dV}{dx} = \frac{x^2 + V^2 x^2}{Vx^2} = \frac{x^2(1 + V^2)}{Vx^2} = \frac{1 + V^2}{V}$$

$$V + x \frac{dV}{dx} = \frac{1}{V} + V$$

$$x \frac{dV}{dx} = \frac{1}{V} \quad \text{Separable}$$

$$\int V dV = \int \frac{1}{x} dx$$

$$\frac{1}{2} V^2 = \ln x + C, \quad y = Vx \quad \text{ie } V = \frac{y}{x}$$

$$\frac{1}{2} \frac{y^2}{x^2} = \ln x + C$$

$$\frac{1}{2} y^2 = x^2 \ln x + Cx$$

Example 3

Use the substitution $y = \frac{V}{x}$ to solve

$$\frac{dy}{dx} + \frac{y}{x} = xy^2, \quad y(2) = 2$$

Solution

$$y = \frac{V}{x} \xrightarrow{\frac{d}{dx}} \frac{dy}{dx} = -\frac{V}{x^2} + \frac{1}{x} \frac{dV}{dx}$$

Substitute:

$$-\frac{V}{x^2} + \frac{1}{x} \frac{dV}{dx} + \frac{V}{x^2} = x \cdot \frac{V^2}{x^2}$$

-3-

$$\frac{1}{x} \frac{dV}{dx} = \frac{V^2}{x}$$

$$\frac{dV}{dx} = V^2 \quad \text{(Separable)}$$

$$\int \frac{1}{V^2} dV = \int dx$$

$$-\frac{1}{V} = x + C, \quad y = \frac{V}{x}$$

$$\therefore V = xy$$

$$-\frac{1}{xy} = x + C$$

$$-1 = x^2 y + Cxy$$

$$\begin{array}{l} \text{Condition} \\ \{ y(2) = 2 \text{ i.e. } x=2 \\ y=2. \end{array}$$

$$-1 = 8 + 4C$$

$$-9 = 4C \quad \therefore C = -\frac{9}{4}$$

Hence

$$-1 = x^2 y - \frac{9}{4} xy$$

$$9xy - 4x^2 y = 4$$

~~9xy - 4x^2 y = 4~~

Example 4

-4-

Use the substitution $x+y=V$, to homogeneous d.e

$$(3x+3y-4)\frac{dy}{dx} = -(x+y)$$

Solution

$$x+y=V$$

$$\text{i.e. } y=V-x \xrightarrow{\frac{d}{dx}} \frac{dy}{dx} = \frac{dV}{dx} - 1$$

Substitute:

$$(3V-4)\left(\frac{dV}{dx} - 1\right) = -V$$

$$\frac{dV}{dx} - 1 = \frac{-V}{3V-4}$$

$$\frac{dV}{dx} = \frac{-V}{3V-4} + 1 = \frac{-V+3V-4}{3V-4} = \frac{2V-4}{3V-4}$$

$$\frac{dV}{dx} = \frac{2V+4}{3V-4}$$

(Separable)

$$\int \frac{3V-4}{2V+4} dV = \int dx$$

$$\begin{array}{r} \frac{3}{2} \overline{) 2V+4} \\ \underline{-(3V+6)} \\ -10 \end{array}$$

Long division

-10 remainder

$$\int \left(\frac{3}{2} - \frac{10}{2V+4} \right) dV = X + C$$

$$\frac{3}{2}V - 5 \int \frac{2}{2V+4} dV = X + C$$

but

$$\frac{3}{2}V - 5 \ln(2V+4) = X + C, \quad V = x+y$$

$$\frac{3}{2}(x+y) - 5 \ln(2x+2y+4) = X + C$$